

Retro Tron Battle — Hardware Test Checklist

Classic Computing 2026 — Celle, 10.–11. Oktober 2026

0. Before touching hardware

[] Confirm server LAN IP for today. All four clients hardcode 192.168.17.138 (HOST\$/ho\$/HO\$) — edit all four .bas files if it differs, re-tokenize/transfer.

[] Start server: python3 server/tron_server.py — confirm startup banner shows "build 12" (or whatever SERVER_BUILD currently is) and TCP 6502 / HTTP 8080.

[] Confirm build artifacts are today's: clients/c64/client.prg, clients/c64/client_wic64.prg — rebuild via petcat if the .bas source changed since they were last tokenized. See disk-images/README.md for ready-made reference images.

[] If testing the photo feature: sudo python -m pyftplib -p 21 from assets/test_ftproot/, PIN MUSTER (or 0001 / 0002 / 4711 / 0815).

1. Joystick test first — every platform, before the game client

Run the standalone diagnostic before touching the actual game client. No server, no networking — just confirms the stick and its port are wired correctly. Once a stick checks out, physically label it (e.g. a tape strip: "ATARI — PORT 1") with which platform/port it was tested and connected to. With three machines and swappable sticks, a mixed-up cable is an easy way to lose an hour chasing a phantom bug.

[] Atari: tools/atari_joystick_test.bas — check both STICK(0) and STICK(1), label the stick.

[] C64: tools/c64_joystick_test.bas — confirm PORT 2 (\$dc00) responds and PORT 1 (\$dc01) stays idle while only touching the stick in port 2 (both game clients read port 2, a deliberate exception — see CLAUDE.md). If port 1 also twitches, that's the keyboard-scan ghosting bug from CLAUDE.md resurfacing — stop and re-check wiring before moving on. Label the stick.

[] CPC: tools/cpc_joystick_test.bas — check JOY(0) responds in all 4 directions, label the stick.

2. Per-platform client status — what's actually unverified

Platform	Last verified on hardware	Watch for
Atari / FujiNet	Real steered crossplay vs. CPC, 2026-09-20	Full match incl. restart — client must use RUN, not GOTO, between matches. NEW since 2026-09-23: server now artificially delays Atari's MOVE (ATARI_LATENCY_EQUALIZER) to match Meatloaf/CPC latency — not yet playtested, check whether Atari still feels faster.
C64 Meatloaf	Case-sensitivity fix verified 2026-09-20 (crossplay vs. CPC)	Join succeeds despite lowercase-mangling; joystick physically in PORT 2. NEW since 2026-09-23 (build 18): instant-START join-response fix, for the case where a second player is matched immediately — not yet re-verified, test two players joining back-to-back.
C64 WiC64	Never tested on real hardware, ever	See section 3 below — today is the first test. Also includes the same instant-START join-response fix as Meatloaf (build 9, 2026-09-23), also unverified.
CPC / M4	Real steered crossplay vs. Atari & Meatloaf, 2026-09-20	IHTTPMEM blocking joystick during calls is expected, not a bug. NEW since 2026-09-23 (build 7): instant-START join-response fix — not yet re-verified, test two players joining back-to-back.

3. C64 WiC64 — special attention (first real-hardware test)

[] Disk has FOTOFIX.C000 alongside the client (clients/c64/wic64-driver/).

[] Boot: lowercase load"fotofix.c000" actually finds the file.

- [] /join and /tick URLs are lowercase — confirm no "ERR UNKNOWN SESSION".
- [] Joystick on PORT 2 with the CLI/RTS stub (BASIC line 46, poke828,88:poke829,96) actually fixes the ghosting.
- [] Watch the temporary debug prints (BASIC lines 250 and 615: raw join/tick response text) — remove once confirmed working, before the event.
- [] The 2026-09-23 instant-START join-response fix (build 9) needs its first hardware check here too, alongside everything else in this section.
- [] Open question: does SYS49152 hold up under rapid repeated /tick polling, vs. the driver's original rare-call design?

4. Cross-cutting checks (single platform)

- [] Play at least 2 consecutive matches per platform, not just one — this project's worst bugs were “works once, fails the second time” (Atari GOTO-vs-RUN, C64 device-path scrambling).
- [] Server display: logos load per match, scroll text readable at the current SCROLL_SPEED, waiting screen cycles cleanly through scroll text -> high-score list -> tribute screen (the fourth view, a fotofix/hans thank-you, was removed 2026-09-23 — confirm it's actually gone from the rotation).
- [] Roughly time how long it feels between moving the stick and the on-screen character turning, per platform — this gives real latency numbers for section 5, instead of guessing. For Atari, this now includes the artificial ATARI_LATENCY_EQUALIZER delay on top of the real (near-zero) network latency.

5. Smoke test: all clients battle each other

The server only ever pairs two players at a time (first two connections) — plan sequential 1v1 matches, not a 4-way free-for-all. Suggested pairing order:

- [] Atari vs C64 (Meatloaf) — both baseline-stable; confirms cross-play/pairing/win-detection itself works before anything experimental joins in.
- [] Atari vs C64 (WiC64) — fastest client against the least-tested one.
- [] C64 (Meatloaf) vs CPC — two HTTP-polling clients against each other.
- [] C64 (WiC64) vs CPC — the two least-proven clients against each other.
- [] Atari vs CPC.
- [] Optional, only with a second physical C64: Meatloaf vs WiC64 directly.

On tick speed: server/tron_server.py runs one fixed-rate game loop (TICK_RATE = 4/sec = 250ms/tick, see the TICK_RATE config block) — movement itself advances at the same rate for both players no matter the platform, so that part is already fair. What differs per platform is how fast a direction change reaches the server: the HTTP-polling clients (Meatloaf, WiC64, CPC) can only submit a new direction once their current blocking /tick round-trip completes (~300ms/~1280ms/~550ms respectively, measured on real hardware/VICE — see CLAUDE.md). Atari keeps a raw TCP socket open and used to send MOVE the instant you steer — **as of 2026-09-23, the server now artificially delays Atari's MOVE too** (ATARI_LATENCY_EQUALIZER, a random delay from ATARI_LATENCY_MIN/MAX) specifically to close that gap. This is new and not yet playtested: use this smoke test to judge whether Atari still feels unfairly fast, and tune the MIN/MAX range (or set ATARI_LATENCY_EQUALIZER = False to fall back to the old instant behavior) based on what you actually feel here — don't tune blindly the night before the event.

If a pairing misbehaves and you need to isolate whether it's the pairing or one specific client, bots/tron_bot.sh can stand in for the TCP side (Atari-protocol) as a predictable opponent.

6. Suggested order for the day

Server smoke test alone -> joystick tests for all three platforms, with labeling (section 1) -> solo client test per platform: C64 Meatloaf -> Atari -> CPC -> C64 WiC64 last (most experimental — if it doesn't pan out today, Meatloaf is the fallback for CC2026, so that decision should get made with time still on the clock) -> once all four are solo-verified, run the cross-platform smoke test pairings from section 5.

Generated 2026-09-23, reflecting the server-side Atari input-latency equalizer and the instant-START join-response fix added that day. See CLAUDE.md, section "Current status", for the up-to-date source of truth — update this checklist with today's findings after testing, so it doesn't drift out of sync again.